

# 22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

## S2 B.Tech CSEAI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

### CASE STUDY

#### PHASE – 2 (Design)

#### Architectural Modelling (CO2)

Group No : 19

Team Members :

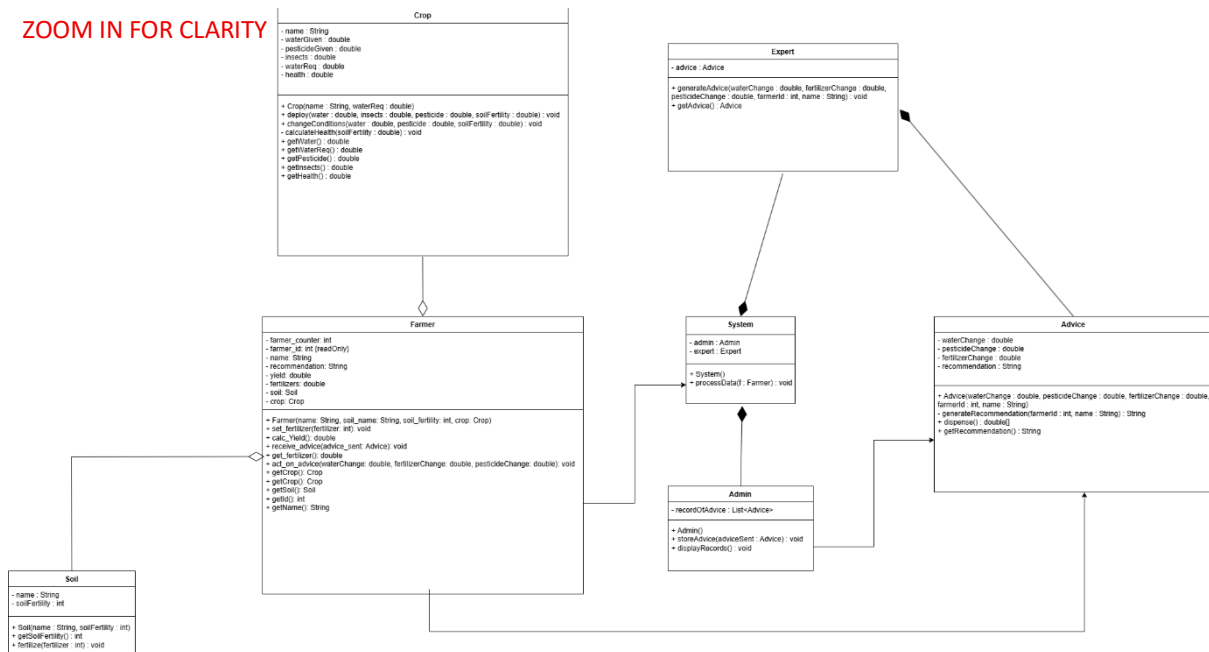
| Roll No          | Student Name                |
|------------------|-----------------------------|
| AM.SC.U4AIE25110 | Chakradhar Sriram Karthik D |
| AM.SC.U4AIE25127 | Tej Vardhan M               |
| AM.SC.U4AIE25142 | Harshadeep Chowdary N       |
| AM.SC.U4AIE25144 | Praveen M                   |

1. Problem Title : Smart Agriculture Monitoring System

### STRUCTURAL DESIGN

2. Class diagram (draw class diagram that shows different classes and their relationships. Use UML notations)

ZOOM IN FOR CLARITY



## **BEHAVIUIORAL DESIGN**

### 3. Use-case diagram

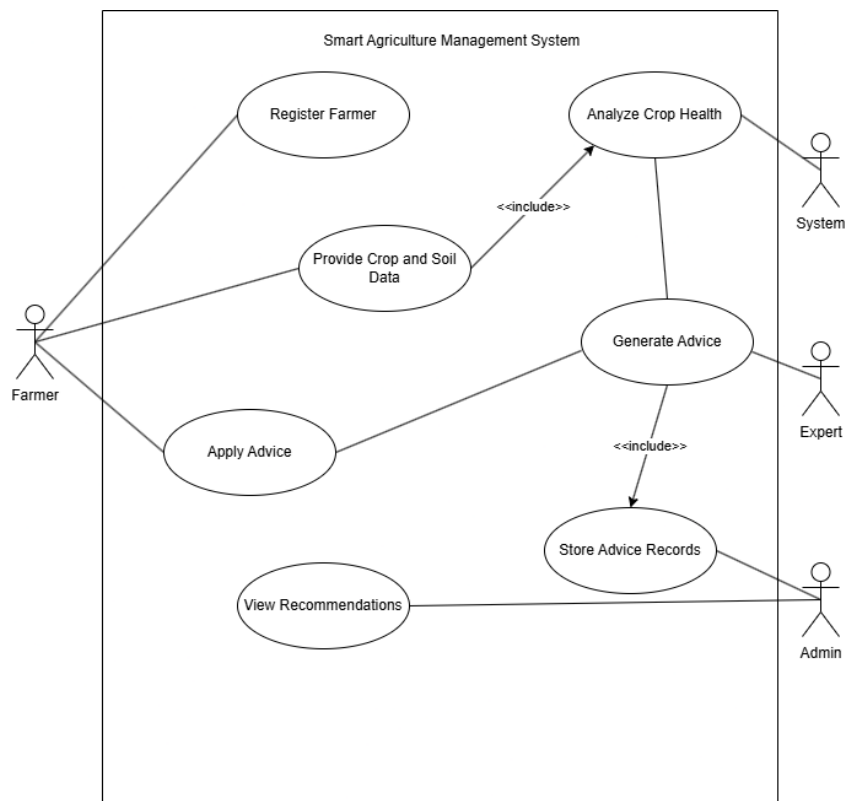
#### a. List of actors

1. Farmer – Inputs data and receives recommendations
2. Expert – Generates advice based on analysis
3. Admin – Stores and manages advice records
4. System – Processes data and performs analysis

#### b. List of use-cases

1. Register Farmer - The farmer enters personal details, soil information, and crop details to create a farmer profile.
2. Provide Crop and Soil Data - The farmer supplies inputs like water, fertilizer, and pesticide conditions.
3. Analyze Crop Health - The system evaluates crop health based on soil fertility, water, and pesticide levels.
4. Generate Advice -The expert generates recommendations based on analyzed data.
5. Apply Advice - Farmer updates farming conditions according to received advice.
6. Store Advice Records - The admin stores all generated advice for future reference.
7. View Recommendations - Farmer views recommendations provided by system.

#### c. Use case diagram (use UML notations)

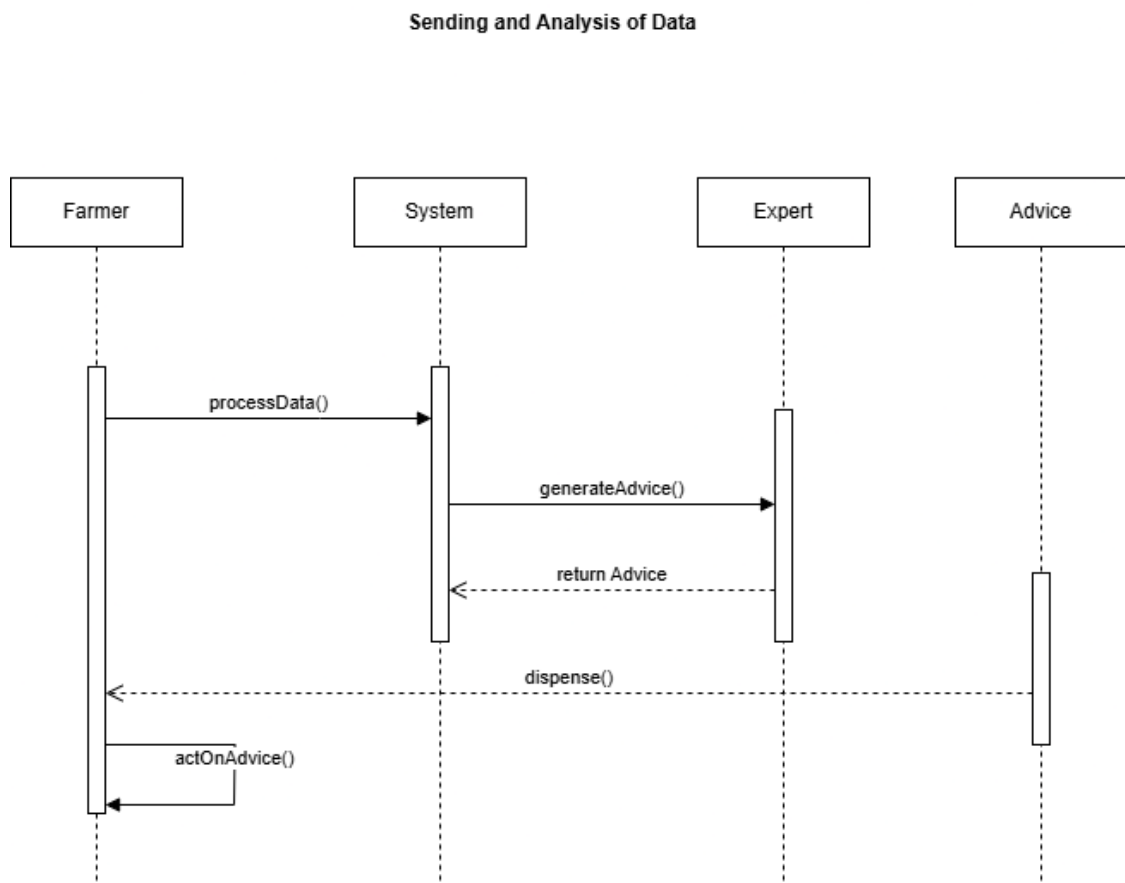


4. Sequence diagram (use UML notations)
- a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

Description:

This use case combines sending data and processing data, where the Farmer initiates the process by providing input to the System. The System analyzes the data and interacts with the Expert to generate appropriate advice, which is then returned and applied by the Farmer.

Diagram:



- b. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

Description:

This use case illustrates the process of storing and viewing recommendations within the system. The System generates advice through the Expert and stores it in the Admin. The Admin then retrieves and displays each recommendation by iterating through the stored Advice objects.

Diagram:

